FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

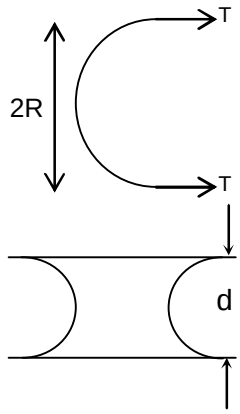
Physics

PART – I

SECTION – A

1. C

Sol. The shape of water layer between the two plates is shown in the figure.



Thickness d of the film = 0.12 mm = 0.012 cm.

Radius R of cylindrical face $= \frac{d}{2}$.

$F = T(2l) = P(l \times 2R)$

$P = \frac{T}{R}$

Pressure difference across the cylindrical surface $= \frac{T}{R} = \frac{2T}{d}$.

Area of each plate wetted by water = A.

Force F required to separate the two plates is given by

F = pressure difference $\times$ area $= \frac{2T}{d} A$

$= \frac{2 \times 75 \times 8}{0.012} = 10^5$ dynes

2. A

Sol. $\left(\text{Loss in K.E. of combined system}\right) = \left(\text{Gain in P.E. of combined system}\right)$

$\Rightarrow \quad \frac{1}{2}(m_1 + m_2)\, V^2 = (m_1 + m_2)\, \frac{gh}{4}$

$\Rightarrow \quad V = \sqrt{2g\left(\frac{h}{4}\right)} \qquad \ldots(1)$

$m_1 gh = \frac{1}{2} m_1 v^2 \quad \Rightarrow \quad v = \sqrt{2gh}$

Further by Law of Conservation of Momentum $(m_1 + m_2)\, V = m_1 v$

$\Rightarrow \quad (m_1 + m_2)\sqrt{\frac{2gh}{4}} = m_1\sqrt{2gh}$

$\Rightarrow \quad \frac{1}{2}(m_1 + m_2) = m_1 \quad \Rightarrow \quad m_1 = m_2$

3. B

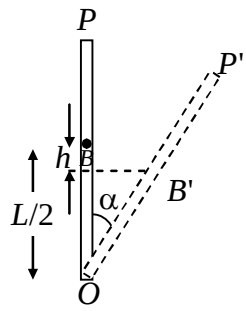
Sol. Consider an elementary part of radius x and width dx.

$$B = \int dB = \int_{R_1}^{R_2} \frac{\mu_0 I}{2\ x} \frac{100}{(R_2 - R_1)}\, dx$$

$$= \frac{2.3 \times 2\pi \times 10^{-7} \times 100}{5 \times 10^{-2}} \ln 2 = 2.003 \times 10^{-4}\ T$$

4. B

Sol. When the rod falls through an angle α the C.G. falls through a height h.



In $\Delta OB'B$,

$$\cos\alpha = \frac{\left(\frac{L}{2} - h\right)}{L/2}$$

$$\text{i.e. } h = \frac{L}{2}(1 - \cos\alpha)$$

K.E. rotation = Decrease in P.E.

$$\text{i.e. } \frac{1}{2} I\omega^2 = mgh$$

$$\text{i.e. } \frac{1}{2}\left(\frac{mL^2}{3}\right)\omega^2 = mg\frac{L}{2}(1 - \cos\alpha) \quad \text{or } \omega = \sqrt{\frac{6g}{L}} \sin\frac{\alpha}{2}$$

5. ABC

Sol. (A) At maximum extension $V_A = V_B = 0$

$$2mgx = \frac{1}{2} kx^2$$

$$x_m = \frac{4mg}{k}$$

(B) $2mg\frac{2mg}{k}=\frac{1}{2}k\left(\frac{2mg}{k}\right)^2+\frac{1}{2}(m+2m)v^2$

$V=2g\sqrt{\frac{2m}{3k}}$

(C) Spring force $\Rightarrow \frac{4mg}{k}\times k=4mg$

net downward force $\Rightarrow$ 2 mg

$a=\frac{4mg-2mg}{3m}=\frac{2}{3}g$

(D) for $x=\frac{2mg}{k}$, a = 0

$\therefore$ T = 2 mg

6. A

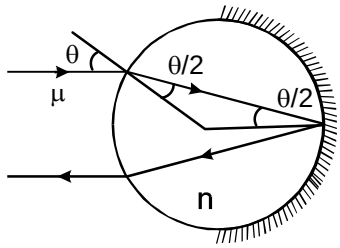
Sol. $50=\frac{T_{Y_2}}{\ln 2}\ln\frac{1}{1-0.01}$

$\Rightarrow T_{Y_2}=\frac{50\ln(2)}{\ln\frac{100}{99}}$

7. A

Sol. $\frac{\sin\theta}{\sin\theta/2}=\frac{\theta}{\theta/2}=\mu$

$\therefore \mu = 2$



8. AD

Sol. Negative of potential gradient is equal to electric field.

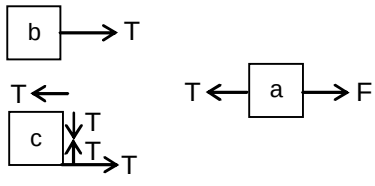
9. BC

Sol. F – T = 10 × a …(1)

T = 10 × a …(2)

F = 20a $\Rightarrow$ a = 1

T = 10 N, $a_a = a_b = 1,\ a_c = 0$



10. ABC

Sol. When switch is closed, all charges will go on outer surface of outer shell and inside E = 0 therefore equipotential at all points 1, 4 and 3 and at point 4 has some potential.

11. C

Sol. $\because \vec{B}$ and $\vec{F}_2 \perp$, $\vec{V}_1 \perp \vec{F}_1$ $\vec{V}_2 \perp \vec{F}_2$

$\therefore$ qa – qc = 0

a = C

12. C

Sol. $\lambda = \frac{d}{10}$, path define = $n\lambda$ [n = $\pm$ 1, + 2 $\pm$ 3]

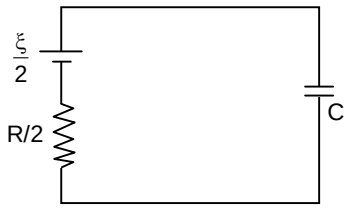
$\therefore \quad |n = \pm 1, \pm 2, \pm 3|$

$\therefore \quad |n| = 6.$

SECTION – C

13. 00000.12

Sol. Equivalent ckt.



$$P_{max} = \left(\frac{\xi}{4}\right)\cdot\frac{\xi/2}{R} = \frac{\xi^2}{8R} = 0.125\,\frac{\xi^2}{R}$$

14. 00000.50

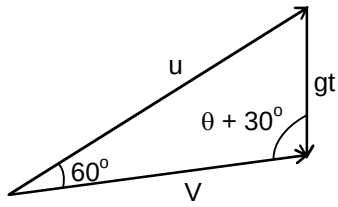
Sol. $$W_3 - W_4 = -\frac{1}{4\pi t_0}\frac{q^2}{r_{34}} = -\frac{1}{4\pi t_0}\frac{q^2}{r2\sin\frac{3\pi}{n}}$$

$$= \frac{-q^2}{8\pi\varepsilon_o r\sin\frac{3\pi}{n}} = 0.5\left[\frac{-q^2}{4\pi t_0 r\sin\left(\frac{3\pi}{n}\right)}\right]$$

15. 00000.86

Sol. $$\frac{gt}{\sin 60^o} = \frac{u}{\sin(30+\theta)}$$

$$t = \frac{\sqrt{3}\,u}{2\,g}\,\frac{1}{\sin(\theta+30^o)}$$



16. 00024.88

Sol. In one oscillation, distance covered = 4 × 3 = 12 cm

Here, $\omega = \frac{2\pi}{T} \Rightarrow T$ = 4s. Now, 8.5 = 4 × 2 + 0.5

$$24 + \left(3 - 3\cos\frac{\pi}{2}\times 0.5\right) = \left(27 - \frac{3}{\sqrt{2}}\right)\text{cm} = (27 - 1.5 \times 1.414)\text{ cm} = 24.88\text{ cm}$$

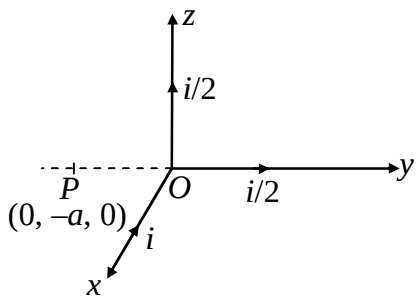
17. 00001.11

Sol. The field at *P* due to current *i* along x-axis is $\frac{\mu_0 i}{4\pi a}\hat{k}$,

and due to the current $\frac{i}{2}$ along z-axis is $\frac{\mu_0 i}{8\pi a}(-\hat{i})$.

The resultant field at *P* is $\frac{\mu_0 i}{4\pi a}\sqrt{1^2+\left(\frac{1}{2}\right)^2} = \frac{\sqrt{5}}{8}\frac{\mu_0 i}{\pi a}$

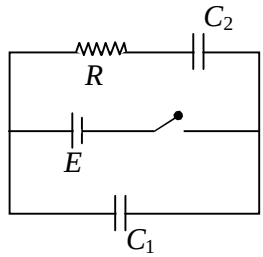
$= \frac{\sqrt{5}}{2}\frac{\mu_0 i}{4\pi a} = 1.118\frac{\mu_0 i}{4\pi a}$



18. 00000.63

Sol. Rearranging the circuit, we observed that C_1 is joined directly to the cell and acquires its full charge when *S* is closed. It plays no part in the charging of C_2 through *R*.

So, $q_2 = Q_0(1-e^{-t/\tau})$ = $\theta_0(1 - 0.37) = 0.63\ \theta_0$



Chemistry

PART – II

SECTION – A

19. C

Sol. NaOH is not used as a dehydration agent.

20. C

Sol.

$$\begin{array}{lccccc} & A(g) & \rightarrow & 4B(g) & + & 3C(g) \\ \text{At } t=0: & P_o & & 0 & & 0 \\ \text{At } t=20\,\text{min}: & P_o-x & & 4x & & 3x \end{array}$$

$P_o - x + 4x + 3x = 400 \Rightarrow P_o + 6x = 400 \Rightarrow 6x = 400 - 100 = 300$

$x = 50$

$$K = \frac{0.693}{t_{1/2}} = \frac{0.693}{20\times 60} = 5.7\times 10^{-4}\ \text{sec}^{-1}$$

21. B

Sol. $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

22. A

Sol. $E_{cell} = E^o - \frac{0.0591}{2}\log\frac{\left[Zn^{2+}\right]}{\left[Cu^{2+}\right]}$

E_{cell} can be increased by diluting the concentration of $[Zn^{2+}]$

23. ABC

Sol. Dipole moment of $NH_3 > NF_3$, $SCl_2 > BeCl_2$(zero value)
$SF_4 > CF_4$(zero value)

24. BC

Sol. It is an endothermic reaction as the equilibrium constant increases with increasing temperature.

25. BCD

Sol. Gr-III, radicals are precipitated with NaOH.

26. CD

Sol. For Ni^{2+} the configuration is $t_{2g}^6 e_g^2$ and Zn^{2+} is $t_{2g}^6 e_g^4$.

27. AC

Sol. $\mu = 2\mu\cos\frac{\theta}{2}$ [When two diploes are identical]

28. CD

Sol. Carbon reduction and distillation(refining) are common for both the ores

$ZnO + C \longrightarrow Zn + CO$

$Zn(\text{impure}) \xrightarrow{\text{distillation}} Zn(\text{pure})$

29. CD

Sol. Na^+ ions are present at octahedral voids, which are body centre and edge centre positions.

30. A

Sol. $CH_3COCH_2CH=CHCH_2COCH_3 \xrightarrow{MnO_4^-/H^+} 2CH_3COCH_2COOH$

$CO_2 + CH_3COCH_3 \xleftarrow{\Delta}$

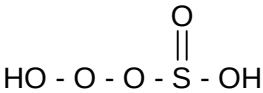
SECTION – C

31. 00001.00

Sol. The valence electrons are $3s^2 3p^1$ for which $\ell = 0$ and 1.

32. 00002.00

Sol. The structure of H_2SO_5 is:



33. 00001.00

Sol. Total charge on four oxide ions = $-2 \times 4 = -8$

Let 'n' no. of $X^{-8/3}$ ions will replace O^{2-} ions

$\therefore\ n \times \frac{-8}{3} = -8 \Rightarrow n = 3$

$\therefore$ One vacancy will be created.

34. 00005.50

Sol. $\Delta T_f = K_f \times n \times \frac{1000}{W}$

Or, $4 = 1.86 \times \frac{6.2}{62} \times \frac{1000}{W}$

$\therefore$ W = 46.5 g

$\therefore$ Mass of water converted to ice = 52 – 46.5 = 5.5

35. 00003.74

Sol. It is a buffer solution

$pH = p^{K_a} + \log \frac{[CH_3COONa]}{[CH_3COOH]} = 4.74 + \log \frac{1}{10} = 3.74$

36. 00012.56

Sol. $\sqrt{\frac{3RT}{M}} = \sqrt{\frac{2RT}{M}}$

or, $\sqrt{\frac{3R \times 500}{30}} = \sqrt{\frac{2R \times 314}{M}}$

On solving, M = 12.56

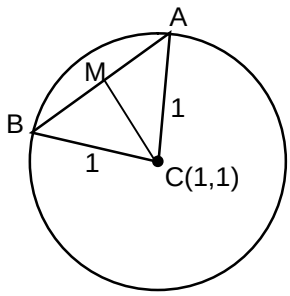
Mathematics

PART – III

SECTION – A

37. A

Sol. The given circle is $(x-1)^2 + (y-1)^2 = 1$.



In the figure $MC = 1 \times \frac{\sqrt{3}}{2}$

$\Rightarrow$ locus of the point M is

$(x-1)^2 + (y-1)^2 = \frac{3}{4}$.

38. A

Sol. We have max.$\{|x-1|, |y-2|\} = 4$.

If $|x-1| \geq |y-2|$ then $|x-1| = 4$,

i.e. if $(x+y-3)(x-y+1) \geq 0$ then $x = -3$ or 5,

If $|y-2| \geq |x-1|$ then $|y-2| = 4$

i.e. $(x+y-3)(x-y+1) \leq 0$, then $y = -2$ or 6.

So, the locus of P bounds a square, the equation of whose sides are

$x = -3$, $x = 5$, $y = -2$, $y = 6$.

Thus the area is $(8)^2 = 64$.

39. C

Sol. a, b > 0 and c < 0

angle between OA and OB is $\frac{\pi}{2}$.

Since z_2 lie in 2nd quadrant

$\Rightarrow \bar{z}_2$ will lie in 3rd quadrant.

$\Rightarrow \sqrt{2} < |z_1 - \bar{z}_2| \leq 2$.

Also true if z_2 lies in 3rd quadrant.

40. B

Sol. Hint : $a_1 = \frac{a}{r}, a_2 = a, a_3 = ar$

Solution : If $a_1, a_2, a_3 (a_1 > 0)$ are in G.P.

$\Rightarrow 9a_1 + 5a_3 \leq 14a_2$

$\Rightarrow 9\frac{a}{r} + 5ar \leq 14a$

$\Rightarrow \frac{9}{r} + 5r \leq 14$

$\Rightarrow 5r^2 - 14r + 9 \leq 0$

$\Rightarrow 5r^2 - 5r - 9r + 9 \leq 0$

$\Rightarrow 5r(r-1) - 9(r-1) > 0$

$\therefore$ The interval is $\left[1, \frac{9}{5}\right]$

41. BD

Sol. Let $I = \int \frac{x\, e^x}{\left(1+e^x\right)} dx$

$$= \int x . \frac{e^x}{\sqrt{\left(1+e^x\right)}} dx$$

$$= x . 2\sqrt{\left(1+e^x\right)} - \int 1 . 2\sqrt{\left(1+e^x\right)} dx$$

$$= 2x\sqrt{\left(1+e^x\right)} - 2\int \sqrt{\left(1+e^x\right)} dx$$

In second integral

Put $1+e^x = t^2$

$\therefore$ $dx = \frac{2t\, dt}{t^2-1}$

Then, $= 2x\sqrt{\left(1+e^x\right)} - 4\int \frac{t^2-1+1}{\left(t^2-1\right)} dt$

$$= 2x\sqrt{\left(1+e^x\right)} - 4\int \left(1 + \frac{1}{t^2-1}\right) dt$$

$$= 2x\sqrt{\left(1+e^x\right)} - 4\left\{t + \frac{1}{2}\ln\left(\frac{t-1}{t+1}\right)\right\} + c$$

$$= 2x\sqrt{\left(1+e^x\right)} - 4\sqrt{\left(1+e^x\right)} - 2\ln\left(\frac{\sqrt{\left(1+e^x\right)}-1}{\sqrt{\left(1+e^x\right)}+1}\right) + c$$

$$= \left(2x-4\right)\sqrt{\left(1+e^x\right)} - 2\ln\left(\frac{\sqrt{\left(1+e^x\right)}-1}{\sqrt{\left(1+e^x\right)}+1}\right) + c$$

On comparing

$$f\left(x\right) = 2x-4,\ g\left(x\right) = \frac{\sqrt{\left(1+e^x\right)}-1}{\sqrt{\left(1+e^x\right)}+1}$$

42. ABCD

Sol. Let $f\left(x\right) = e^x - x^e$

$f'\left(x\right) = e^x - ex^{e-1} = 0$ if $e^x = ex^{e-1}$

$\Rightarrow e^{x-1} = x^{e-1} \rightarrow x = e$

If $x > e,\ f'\left(x\right) > 0 \Rightarrow f\left(x\right) > f\left(e\right)$

If $x < e,\ f'\left(x\right) < 0,\ f\left(x\right) > f\left(e\right)$

as $\frac{\pi}{2} < e,\ e^{\frac{\pi}{2}} > \left(\frac{\pi}{2}\right)^e$

as $\pi > e,\ e^{\pi} > \pi^{e}$

as $\pi > e,\ e^{\pi} > \pi^{e}$

as $3 > e,\ e^{3} > 3^{e}$

as $2 < e,\ e^{2} > 2^{e}$

43. BD

Sol. $^{100}C_6 + 4.\ ^{100}C_7 + 6.\ ^{100}C_8 + 4.\ ^{100}C_9 + {}^{100}C_{10}$

$= {}^{4}C_0.\ ^{100}C_6 + {}^{4}C_1.\ ^{100}C_7 + {}^{4}C_2\ ^{100}C_8 + {}^{4}C_3.\ ^{100}C_9.\ ^{100}C_9 + {}^{4}C_4\ .\ ^{100}C_{10}$

$= {}^{4}C_0.\ ^{100}C_{94} + {}^{4}C_1 \times {}^{100}C_{93} + {}^{4}C_2 \times {}^{100}C_{92} + {}^{4}C_3 \times {}^{100}C_{91} + {}^{4}C_4 \times {}^{100}C_{90}$

$= {}^{104}C_{94} = {}^{104}C_{10}$

$\Rightarrow 104 + 94 = 198,\ 104 + 10 = 114$

44. AC

Sol. Take $\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right] = 0$

$\Rightarrow \beta = 3$

$\therefore \vec{c} = (\alpha + 1)\hat{i} + 2\hat{j} + \hat{k}$

$\left|\vec{c}\right| = \sqrt{6}$

$(\alpha + 1)^2 + 4 + 1 = 6 \qquad \Rightarrow \alpha = 0, -2$

$\therefore\ \alpha + \beta = 3, 1$

45. ABC

Sol. $\because |A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix} = 3(-3+4) - 2(-3+4) + 0 = 1$

$\because$ $\text{adj}(\text{adj}\,A) = |A|^{3-2} A = A$ and $|\text{adj}(\text{adj}\ A)| = |A| = 1$

Also, $|\text{adj}\ A| = |A|^{3-1} = |A|^2 = 1^2 = 1$

46. AD

Sol. $\because \sin B = \dfrac{b \sin A}{a}, \sin B < 1$

Then, $\dfrac{b \sin A}{a} < 1 \Rightarrow \sin A < \dfrac{a}{b} \Rightarrow$ For triangle to be possible a < b

$\therefore$ $b \sin A < a$

If $b \sin A = a \Rightarrow \angle B = \dfrac{\pi}{2} \Rightarrow \angle A < \dfrac{\pi}{2}$

47. ABCD

Sol. $\because$ $I_n = \int_0^{\pi/4} \tan^{n-2} x \left(\sec^2 x - 1\right) dx$

$= \int_0^{\pi/4} \tan^{n-2} x . \sec^2 x\, dx - I_{n-2}$

or $I_n + I_{n-2} = \frac{1}{(n-1)}$

For $n = 5,\ I_5 + I_3 = \frac{1}{4}$ …(i)

For $n = 8,\ I_8 + I_6 = \frac{1}{7}$ …(ii)

For $n = 10, I_{10} + I_8 = \frac{1}{9}$ ….(iii)

Subtracting Eq. (iii) from (ii), then

$I_6 - I_{10} = \frac{2}{63}$ ….(iv)

Adding Eqs. (ii) and (iii), we get $I_6 + 2I_8 + I_{10} = \frac{16}{63}$

48. ABCD

Sol. $A_1 = \pi r_1^2;\ A_2 = \pi r_2^2;\ A_3 = \pi r_3^2$

$\sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi}(r_1 + r_2 + r_3) = \sqrt{\pi}(4R + r)$

$\frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{1}{\sqrt{\pi}}\left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}\right) = \frac{1}{\sqrt{\pi}}\left(\frac{1}{r}\right) = \frac{1}{\sqrt{\pi r^2}} = \frac{1}{A}$

$\frac{s^2}{\sqrt{\pi}\, r_1 r_2 r_3} = \frac{s.s(s-c)(s-b)(s-c)}{\sqrt{\pi}\,\Delta^3} = \frac{s\Delta^2}{\sqrt{\pi}\,\Delta^2} = \frac{1}{A}$

SECTION – C

49. 00004.50

Sol. R cos A = $\frac{3}{\sqrt{5}}$ $\Rightarrow$ – R cos B cos C + R sin B sin C = $\frac{3}{\sqrt{5}}$ and 2R cos B cos C = $\frac{3}{\sqrt{5}}$

$\Rightarrow$ R sin B sin C = $\frac{3}{\sqrt{5}} + \frac{3}{2\sqrt{5}} = \frac{9}{2\sqrt{5}}$

sin B sin C = $\frac{9}{2\sqrt{5}} \times \sqrt{\frac{5}{61}} = \frac{9}{2\sqrt{61}}$.

50. 00000.50

Sol. For $n \in N,\ n < \sqrt{n^2 + n + 1} < n + 1$

$\text{So required answer} = \lim_{n\to\infty}\left(\sqrt{n^2+n+1} - \left[\sqrt{n^2+n+1}\right]\right)$

$= \lim_{n\to\infty}\left(\sqrt{n^2+n+1} - n\right) = \lim_{n\to\infty}\frac{n+1}{\sqrt{n^2+n+1}+n} = \frac{1}{2}$

51. 00006.40

Sol. $\frac{\frac{d^3y}{dx^3}}{\frac{d^2y}{dx^2}} = 8$

$\Rightarrow \quad \ln\frac{d^2y}{dx^2} = 8x + c$

or $\quad \ln y_2 = 8x + c$

Putting x = 0, we have $c = 0$.

$\therefore \quad \ln y_2 = 8x \Rightarrow y_2 = e^{8x}$

$\therefore \quad y_1 = \frac{e^{8x}}{8} + D$

Again putting $x = 0,\ D = -\frac{1}{8}$.

$\Rightarrow \quad y_1 = \frac{e^{8x}}{8} - \frac{1}{8}$

$\Rightarrow \quad y = \frac{e^{8x}}{64} - \frac{x}{8} + E$

Putting $x = 0,\ E = \frac{1}{8} - \frac{1}{64} = \frac{7}{64}$.

Hence, $y = \frac{e^{8x}}{64} - \frac{x}{8} + \frac{7}{64} = \frac{e^{8x} - 8x + 7}{64}$.

52. 00001.60

Sol. Let $I_i = \frac{\left(2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1\right)}{\left(x^2 + 2\right)}dx$

$= \int_{-\sqrt{2}}^{\sqrt{2}} \frac{\left(2x^7 - 10x^5 - 7x^3 + x\right)}{\left(x^2 + 2\right)}dx + \int_{-\sqrt{2}}^{\sqrt{2}} \frac{\left(3x^6 - 12x^2 + 1\right)}{\left(x^2 + 2\right)}dx$

$= 0 + 2\int_0^{\sqrt{2}} \frac{3x^2\left(x^4 - 4\right) + 1}{\left(x^2 + 2\right)}dx$

$= 2\int_0^{\sqrt{2}} \left(3x^2\left(x^2 - 2\right) + \frac{1}{x^2 + 2}\right)dx$

$= 2\left\{\frac{3x^5}{5} - 2x^3 + \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right)\right\}_0^{\sqrt{2}}$

$= 2\left\{\frac{12\sqrt{2}}{5} - 4\sqrt{2} + \frac{1}{\sqrt{2}}.\frac{\pi}{4}\right\}$

$= \frac{24\sqrt{2}}{5} - 8\sqrt{2} + \sqrt{2}.\frac{\pi}{4} = \frac{\sqrt{2}}{20}(5\pi + 64)$

$\therefore k^2 = 64 \Rightarrow k = 8$

53. 00005.25

Sol. $\left|\vec{a}\right| = 1,\ \left|\vec{b}\right| = 1,\ \left|\vec{a} + \vec{b}\right| = \sqrt{3}$

$3 = 1 + 1 + 2\vec{a}.\vec{b} \Rightarrow \vec{a}.\vec{b} = \frac{1}{2}$

$$\vec{C} = \vec{a} + 2\vec{b} - 3\left(\vec{a} \times \vec{b}\right)$$

$$p = \left(\vec{a} \times \vec{b}\right) \times \vec{c} \Rightarrow \left(\vec{a} \times \vec{b}\right) \times \left(\vec{a} + 2\vec{b} - 3\left(\vec{a} \times \vec{b}\right)\right)$$

$$= \left(\vec{a} \times \vec{b}\right) \times \vec{a} + 2\left(\vec{a} \times \vec{b}\right) \times \vec{b}$$

$$= \left(\vec{a}.\vec{a}\right)\vec{b} - \left(\vec{b}.\vec{a}\right)\vec{a} + 2\left(\left(\vec{a}.\vec{b}\right)\vec{b} - \left(\vec{b}.\vec{b}\right).\vec{a}\right)$$

$$= \vec{b} - \frac{1}{2}\vec{a} + 2 \times \left(\frac{1}{2}\vec{b} - \vec{a}\right) = 2\vec{b} - \frac{5}{2}\vec{a}$$

$$\left|\vec{p}\right| = \sqrt{4 + \frac{25}{4} + 10\vec{a}.\vec{b}} = \sqrt{\frac{41}{4} - 5} = \frac{\sqrt{21}}{2}$$

54. 00002.25

Sol. $a^2 + b^2 + c^2 = ab + bc + ca \Rightarrow \frac{1}{2}\left\{(a-b)^2 + (b-c)^2 + (c-a)^2\right\} = 0$

$\Rightarrow a = b = c \Rightarrow A = B = C = 60^o$